

The six-sphere in eighteen pictures

a walkthrough of a Lean development, and a machine-checked proof
that the walkthrough is complete

Abstract

This repository reviews a Lean 4 development that proves, from its own construction, that the six-sphere carries a complex structure, and it adds the several-complex-variables and singular-homology theory that the review needed. A series of eighteen pictures, emitted by Lean itself, walks through that argument from the question to the audit.

This document presents the series, and states and explains the accompanying formal result: the series covers the walkthrough. Every step of the argument has exactly one picture and every picture has a step; every chapter is represented; each picture displays the heading of its step and its own position in a series of eighteen; every layer the argument assumes is displayed, by name, in the picture assigned to it; and this document reproduces the series plate for plate. Those statements are theorems of `Figures/Coverage.lean` and `Figures/Paper.lean`, decided by the Lean kernel from the figure sources themselves.

The coverage result is about the pictures, not about the mathematics they depict: it says the walkthrough has no gaps, not that the proof walked through is correct. What backs the mathematical claim is the development and the audit described in Section 2 and drawn in plate 17.

1 What this document is

The pictures are the primary artefact. They are written by Lean: `Figures/Svg.lean` is a small SVG writer that depends on nothing but core Lean, `Figures/Draw.lean` is the drawing vocabulary, the modules under `Figures/Scenes/` define one scene per step, and elaborating `Figures/Emit.lean` writes every picture to `figures/svg/`. The plates at the end of this document are those files, converted to PDF.

The document has two halves. The first, up to Section 5, says what the series is, what it means for it to cover the argument, and what is proved about that. The second is the series itself, one plate to a page, in order.

Nothing in the figure library is used by any theorem of the repository. The library only draws, and the coverage theorems only inspect what it draws.

2 The development the series walks through

The target proposition is that the six-sphere admits an atlas modelled on a three-dimensional complex vector space whose transition maps are complex analytic. The development obtains it in four stages:

1. a compact complex threefold X is constructed, and the facts about it that the rest of the proof uses are isolated as a short interface (plates 4-6);
2. X is shown to be simply connected and to have the homology of the six-sphere, hence to be homotopy equivalent to it (plates 7-9);
3. Morse theory recognises a closed smooth six-manifold homotopy equivalent to the six-sphere as homeomorphic to it (plates 10-11);
4. the holomorphic atlas of X is transported along that homeomorphism, and its model space is changed to the one the statement names (plates 12-13).

The facts the comparison assumes from the extracted development are named, one by one, in `submission/HopfMissing/Layers.lean`; they are the *layers*, and they are listed in Table 2. Around that spine this repository adds its own several-complex-variables development, its own singular-homology development, and a verification apparatus: a source scan, a clean build, a whole-environment axiom audit, and a check that every identifier cited in prose resolves. Plates 15-17 draw those three additions.

3 The series, and what it covers

Table 1 is the series: eighteen steps, grouped into chapters, and the file each one is drawn in. It is generated from `Figures/Coverage.lean`, so it cannot drift from the statements below.

4 What is proved

4.1 Reading a picture in Lean

A picture is a value of type `Figures.Svg.Doc`: a size, a background and a tree of nodes. `Figures/Coverage.lean` extracts the text nodes of such a tree,

```
def texts (d : Doc) : List String := (d.nodes.toList.map (nodeTexts depth)).flatten

def shows (d : Doc) (s : String) : Bool :=
  (texts d).any fun t => charsInfix s.toList t.toList

def showsText (d : Doc) (s : String) : Bool := (texts d).contains s
```

so that “the picture displays s ” is a decidable proposition about the picture that the build actually emits. The extraction is written to be reducible by the kernel, and every coverage statement below is closed by `decide`: no statement about the series is asserted in prose and then trusted.

4.2 The steps, and the pictures

The eighteen steps are an inductive type, and `Step.figure` assigns each one a file stem. The catalogue emitted by the build is filed under exactly the eighteen documented stems, in order, and the assignment is a bijection onto them:

```
theorem catalogue_stems : catalogue.toList.map Prod.fst = stems := rfl

theorem figure_mem_stems (s : Step) : s.figure ∈ stems

theorem figure_injective (a b : Step) : a.figure = b.figure → a = b

theorem stem_has_step : ∀ n ∈ stems, ∃ s ∈ Step.all, s.figure = n
```

Together these say that no step is unillustrated, that two steps never share a picture, and that the series contains no picture that illustrates nothing. The steps are also grouped into chapters, and every chapter is represented:

```
theorem chapter_covered : ∀ c ∈ Chapter.all, ∃ s ∈ Step.all, s.chapter = c
```

chapter	step	figure
framing	PROLOGUE — THE QUESTION	01-question
framing	THE STATEMENT	02-statement
framing	THE ROADMAP	03-roadmap
stage 1	STEP 1a — THE CONSTRUCTION	04-threefold
stage 1	STEP 1b — THE COMPLEX STRUCTURE	05-atlas
stage 1	STEP 1c — THE INTERFACE	06-interface
stage 2	STEP 2a — THE FUNDAMENTAL GROUP	07-fundamental-group
stage 2	STEP 2b — THE HOMOLOGY	08-homology
stage 2	STEP 2c — THE HOMOTOPY EQUIVALENCE	09-homotopy-equivalence
stage 3	STEP 3a — MORSE THEORY	10-morse
stage 3	STEP 3b — THE RECOGNITION	11-recognition
stage 4	STEP 4a — TRANSPORT	12-transport
stage 4	STEP 4b — THE MODEL SPACE	13-model-change
conclusion	THE CONCLUSION	14-conclusion
this repository	WHAT THIS REPOSITORY ADDS (1)	15-several-complex-variables
this repository	WHAT THIS REPOSITORY ADDS (2)	16-homology-development
this repository	HOW IT IS CHECKED	17-verification
overview	THE WHOLE ARGUMENT	18-whole-argument

Table 1: The eighteen steps of the walkthrough, and the picture that shows each one. Generated from the Lean sources.

4.3 The assignment is not just a naming convention

A map from steps to file names would prove little on its own: it is a table someone typed. What ties it to the drawings is that each picture is required to display the heading of the step it is assigned to, and its own position in the series:

```
theorem headings_shown :
  Step.all.all (fun s => showsText (docOf s.figure) s.heading) = true

theorem numbering_consistent : numberedWith captions catalogue.toList = true
```

`numberedWith` walks the eighteen captions and the catalogue in step, and is false if the two have different lengths, so a picture added, removed or reordered without renumbering breaks the build. It caught one such defect: before this document was written, plate 1 read “Figure 01 of 17”.

4.4 Every assumption of the argument is on a plate

The layers are what the comparison assumes from the extracted development, and they are the natural measure of whether the walkthrough is complete: a picture series that omitted one would be hiding an assumption. Table 2 lists

every one of them, the name the picture prints for it, and the plate it is printed on, and

```
theorem layers_shown : layerTable.all (fun r => shows (docOf r.figure) r.cited) = true
```

proves that each of those names really is drawn on that plate. Some plates cite a layer by the alias of `submission/HopfMissing/Layers.lean` and some by the declaration that alias abbreviates; the table records both, and `tools/check_layer_coverage.py` checks the remaining, textual half of the claim — that the table lists every layer of that module, and that no row cites a name unrelated to its layer.

4.5 This document reproduces the series

Finally, the plates at the end are generated from the same table. Each plate records the stem of its picture and the three strings that picture draws, and

```
theorem paperCoversSeries :
  plates.map Plate.stem = stems ∧
  (∀ e ∈ catalogue.toList, ∃ p ∈ plates, p.stem = e.1) ∧
  (∀ p ∈ plates, ∃ d : Doc, (p.stem, d) ∈ catalogue.toList) ∧
  plates.all (fun p => showsText (docOf p.stem) p.heading) = true ∧
```

layer	cited in the picture as	figure
layer5_isManifold	space_isManifold	06-interface
layer5_compact	space_compact	06-interface
layer5_t2Space	space_t2Space	06-interface
layer5_secondCountable	space_secondCountable	06-interface
layer5_nonempty	space_nonempty	06-interface
layer5_connected	space_connected	06-interface
layer5_real_dimension	real_dimension	06-interface
layer6_simplyConnected	layer6_simplyConnected	07-fundamental-group
layer7_homology_vanishes_above_six	layer7_homology_vanishes_above_six	08-homology
layer7_homologyTwo_subsingleton	layer7_homologyTwo_subsingleton	08-homology
layer8_homotopyEquiv	layer8_homotopyEquiv	09-homotopy-equivalence
layer9_homeomorphic_of_homotopy	Smale.homeomorphic_sixSphere_of_homotopySixSphere	11-recognition
layer2_sphere_homology_subsingleton	layer2_sphere_homology_subsingleton	08-homology

Table 2: Every layer the argument assumes, and where the series displays it. Generated from the Lean sources.

```
plates.all (fun p => showsText (docOf p.stem) p.subtitle) = true ∧
plates.all (fun p => showsText (docOf p.stem) p.caption) = true
```

proves that the paper has one plate per picture and one picture per plate, in the order of the catalogue, and that the heading, subtitle and caption printed under each plate are the ones its picture draws. The LaTeX for the plates and for the two tables above is written by `Figures/Paper.lean` into `paper/generated/`; only the prose is written by hand.

4.6 The single statement

`Figures/Coverage.lean` collects the series-level statements into one theorem, `seriesComplete`, and `Figures/Paper.lean` collects the document-level ones into `paperCoversSeries`. Both are proved without sorry; the repository’s axiom audit covers the figure library along with everything else.

5 What the coverage proof does not say

It is worth being exact about the scope, since “we covered the whole visualization” is easy to overstate.

- It is a statement about pictures. It says the series is complete and internally consistent as a walkthrough; it says nothing about whether the proof walked through is correct. That is the business of the development and of the audit.
- “The picture displays s ” means some text node of the picture contains s as a substring. It does not mean the string is legible, well placed, or true.

- The steps and the chapters are a chosen decomposition of the argument. The proof that the series covers them is exact; the judgement that they are the right decomposition is not formal. The layer table is the part of the decomposition that is pinned to the development rather than chosen, which is why it carries most of the weight.
- The prose of this document, including the one-line note under each plate, is not checked against anything.
- Several plates are schematic — the threefold as a family of tori, the Morse landscape, the two discs. They show the shape of a construction, not its details.

6 Reproducing this document

From the repository root:

```
lake build Figures      # checks the coverage theorems; rewrites figures/svg/
                        # and paper/generated/ as a side effect
lake exe figures        # the same writes, on demand
sh tools/build_paper.sh # figures/svg -> figures/pdf, then XeLaTeX
```

`tools/build_paper.sh` needs XeLaTeX and one of CairoSVG, rsvg-convert or Inkscape. The rasterised PNGs of the same pictures, for reading without LaTeX, are in `figures/png/`.

The plates

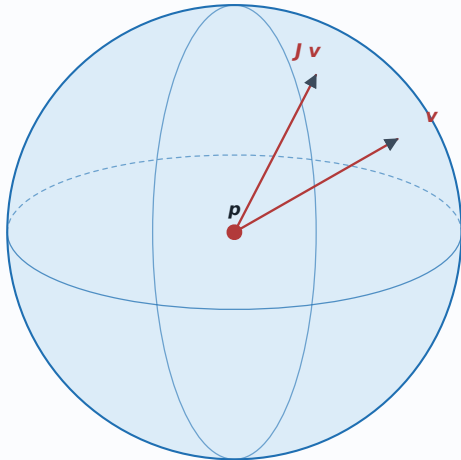
The eighteen pictures of the series, in order, each with the heading and subtitle it draws and the caption in its footer. The pictures are 1600 by 1000 user units; they are reproduced here at the width of the text.

PROLOGUE — THE QUESTION Does the six-sphere carry a complex structure?

PROLOGUE — THE QUESTION

Does the six-sphere carry a complex structure?

Hopf's problem · MathOverflow 1973



$$S^6 = \{ x \in \mathbb{R}^7 : \|x\| = 1 \}$$

with its standard topology as a subspace of $\mathbb{R}^7$

real dimension 6

The classical picture

S^2 and S^6 are the only spheres that admit an almost complex structure $J : TS \rightarrow TS$, $J^2 = -1$. On S^6 one comes from octonion multiplication — but that J is not integrable: it is not induced by holomorphic charts.

Hopf's question (1940s)

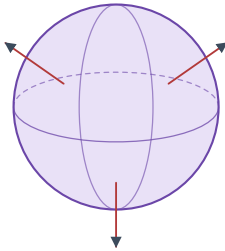
Is there any complex structure on S^6 at all? Equivalently: is there an atlas of S^6 by open subsets of $\mathbb{C}^3$ whose transition maps are holomorphic?

What this repository does

It checks a Lean 4 development that answers YES, re-proves the statement from an extracted copy, audits every axiom used, and adds the several-complex-variables theory the comparison needs. These pictures walk through that proof.

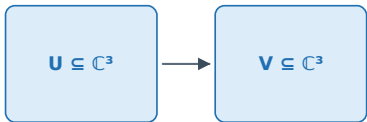
The two structures, side by side

almost complex



a field of rotations by 90° on each tangent space — always exists on S^6

complex (integrable)



charts into $\mathbb{C}^3$ whose transition maps are holomorphic — this is what is at stake

the question: does the second exist?

The answer, as formalised: yes

Figures 02 to 14 walk through the machine-checked proof; figures 15 to 18 cover what this repository adds to it and how the claim is audited.

Figure 01 of 18 · the question

$S^6 \subset \mathbb{R}^7$ · charts modelled on $\mathbb{C}^3$

Hopf's question, and what an integrable complex structure would be.

01-question Figure 01 of 18 · the question

THE STATEMENT What is to be proved, in Lean

THE STATEMENT

What is to be proved, in Lean

the challenge proposition and its two forms

Challenge.lean — the proposition (model-with-corners notation spelled out)

```
theorem mathoverflow_1973 :
  ∃ atlas : ChartedSpace (EuclideanSpace ℂ (Fin 3)) (unitSphere 6),
    let I := atlas
    IsManifold (modelWithCornersSelf ℂ (EuclideanSpace ℂ (Fin 3))) 1
      (unitSphere 6)

where unitSphere 6 = Metric.sphere (0 : EuclideanSpace ℝ (Fin 7)) 1
```

Nothing here is vacuous

S^6 is nonempty and compact, so no single chart can cover it: the existential really ranges over atlases with at least two charts and holomorphic transitions. The two other formulations used in this repository agree with it as statements, and the agreement is itself machine-checked.

the space

$\text{unitSphere } 6$ is S^6 with its standard subspace topology from $\mathbb{R}^7$ — no new topology is introduced, so the atlas must be compatible with it.

the atlas

$\text{ChartedSpace } H \ M$: charts are partial homeomorphisms onto open subsets of $H = \mathbb{C}^3$, and every point lies in the source of one of them.

the regularity

$\text{IsManifold } \dots \ 1$ asks the transition maps to be $\text{ContDiffOn } \mathbb{C} \ 1$ — differentiable over $\mathbb{C}$, hence holomorphic. It is not merely real C^1 .

The two forms that get proved

$S6HasComplexStructure$ — regularity T (analytic): the form stated by this project;
 mathoverflow_1973 — regularity 1 : the challenge form. The analytic form is the stronger of the two, and it is the one the construction delivers.

Read the statement as: there is an atlas of S^6 by open subsets of $\mathbb{C}^3$ whose transition maps are holomorphic.

That is the Hopf problem. Everything that follows is the construction of such an atlas, and the proof that it is one.

Where it lives in this repository

$\text{HopfMissing.S6HasComplexStructure}$ (the statement) · $\text{HopfMissing.s6_hasComplexStructure}$ (analytic form, regularity T) · $\text{HopfMissing.s6_hasComplexStructure_contDiff_one}$ (C^1 form)

Figure 02 of 18 · the statement

$S^6 \subset \mathbb{R}^7$ · charts modelled on $\mathbb{C}^3$

The formal statement dissected, and why it is not vacuous.

02-statement Figure 02 of 18 · the statement

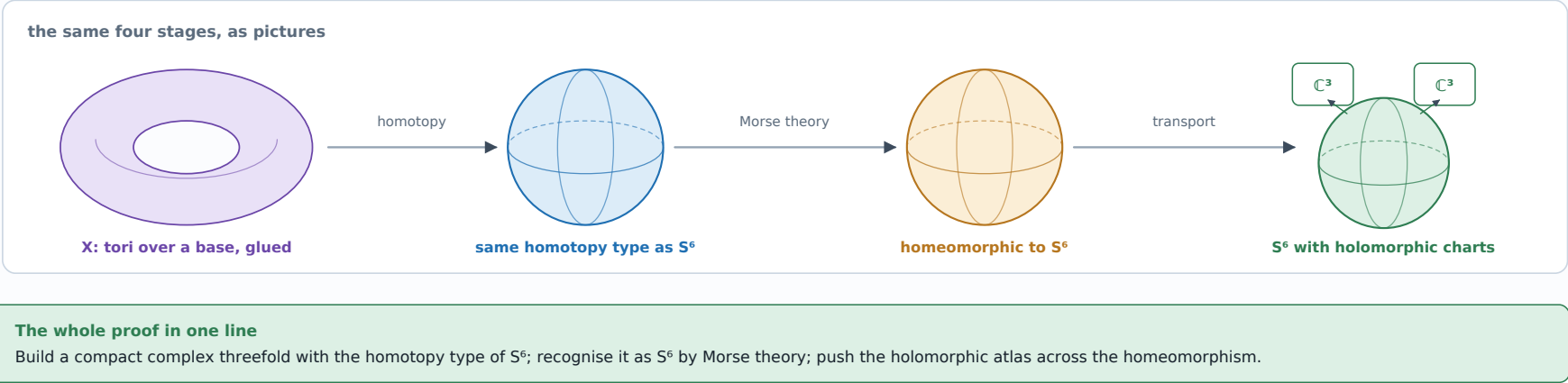
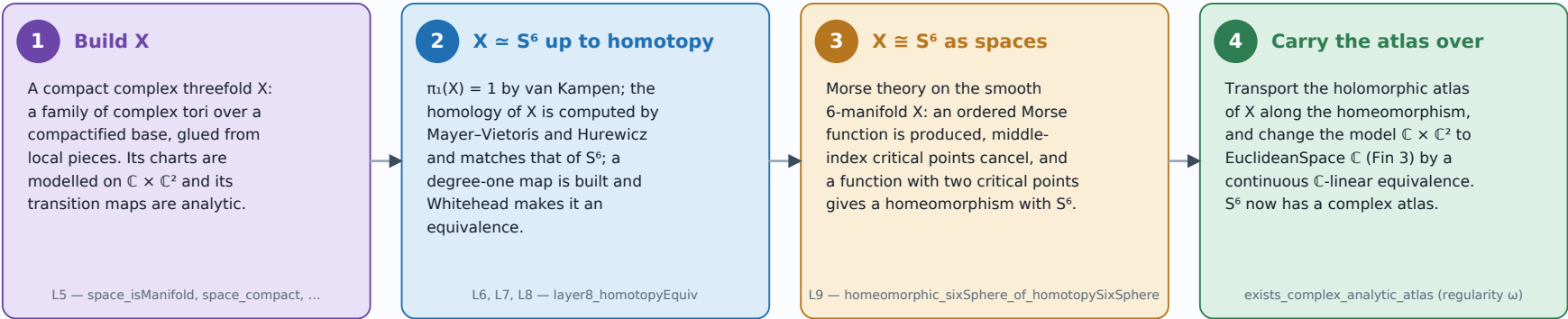
THE ROADMAP How the complex structure on S^6 is obtained

THE ROADMAP

How the complex structure on S^6 is obtained

four stages, and the layers each one settles

The idea: do not build charts on S^6 directly. Build a compact complex threefold X , then show that X is S^6 as a topological space, and carry the charts across.



STEP 1a — THE CONSTRUCTION A compact complex threefold X

STEP 1a — THE CONSTRUCTION

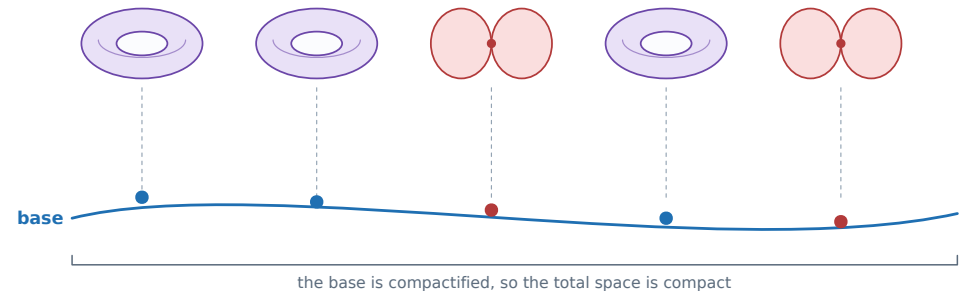
A compact complex threefold X

SpecialPeriods.Threefold.Space

schematic: complex 2-tori varying over a compactified base curve

generic fibre: a complex 2-torus

special fibres



X = the total space of this family, glued from the local pieces

complex dimension 3 — one parameter along the base, two along the fibre

real dimension 6 — the same as S^6 , which is what makes the comparison possible

How X is built

Local pieces are described by explicit period data and glued over a compactified parameter space. Each piece is an open subset of $\mathbb{C} \times \mathbb{C}^2$, so the glued space comes with charts into $\mathbb{C} \times \mathbb{C}^2$, and the gluing maps are holomorphic where they overlap. Everything below is proved about that glued space.

Why a threefold at all

A complex structure on S^6 cannot be written down directly. Instead one builds a complex manifold whose underlying topological space can be recognised as S^6 . The complex geometry is done once, on X , and then carried across.

What step 1 delivers

A space `Space` with a `ChartedSpace` structure modelled on $\mathbb{C} \times \mathbb{C}^2$, whose transition maps are complex analytic, and which is compact, Hausdorff, second countable, connected and nonempty.

The Lean names

`SpecialPeriods.Threefold.Space` · `space_isManifold` · `space_compact` · `space_t2Space` · `space_secondCountable` · `space_connected` · `space_nonempty` · `real_dimension`

Figure 04 of 18 · step 1a, the construction

layer L5 of the development

The threefold as a family of complex tori over a compactified base.

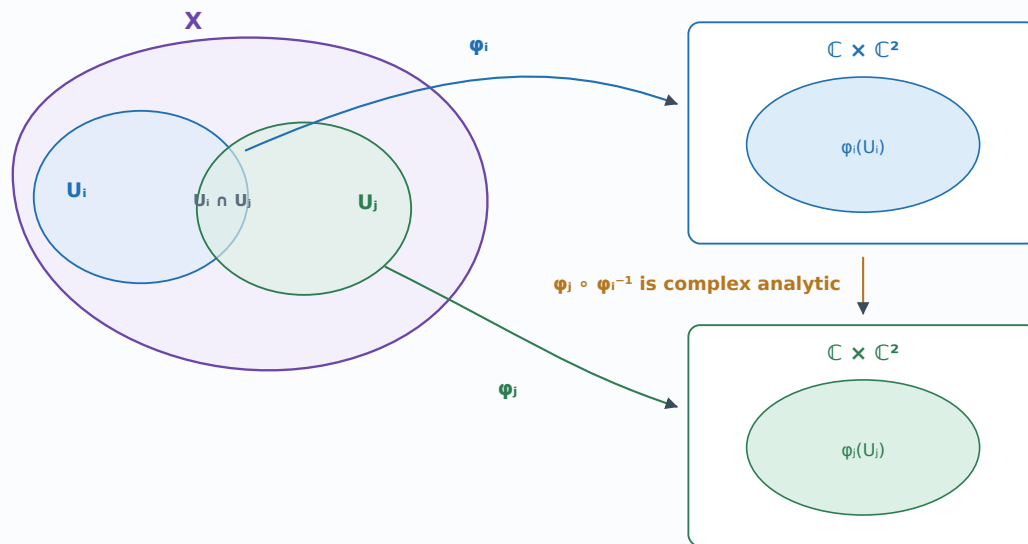
04-threefold Figure 04 of 18 · step 1a, the construction

STEP 1b — THE COMPLEX STRUCTURE Charts on X , and why the transitions are holomorphic

STEP 1b — THE COMPLEX STRUCTURE

Charts on X , and why the transitions are holomorphic

IsManifold (model $\mathbb{C} \times \mathbb{C}^2$) ω Space



What the atlas condition says

A ChartedSpace is an atlas: partial homeomorphisms ϕ_i from open subsets of X onto open subsets of the model, covering X . IsManifold adds that every composite $\phi_j \circ \phi_i^{-1}$ lies in the analytic groupoid of the model — it is holomorphic, with holomorphic inverse, on the image of the overlap.

Regularity ω , not just 1

space_isManifold is proved with regularity ω (complex analytic), which is strictly stronger than the C^1 the challenge statement asks for. Holomorphy is what the gluing construction gives, so nothing is lost by asking for it.

The point of this step

X is now a complex manifold. Steps 2 and 3 forget the complex structure and study only the space; step 4 puts it back.

The Lean names

SpecialPeriods.Threefold.space_isManifold · contDiffGroupoid · HasGroupoid · ChartedSpace

Figure 05 of 18 · step 1b, the complex structure

layer L5 of the development

Charts into the model space, and why the transitions are holomorphic.

05-atlas Figure 05 of 18 · step 1b, the complex structure

STEP 1c — THE INTERFACE Exactly what is known about X when step 2 begins

STEP 1c — THE INTERFACE

Exactly what is known about X when step 2 begins

the layer-5 facts, and the names that carry them

property of X	what it says	Lean name
complex manifold	charts into $\mathbb{C} \times \mathbb{C}^2$, analytic transitions	space_isManifold
compact	every open cover has a finite subcover	space_compact
Hausdorff	points are separated by open sets	space_t2Space
second countable	a countable base of open sets	space_secondCountable
connected	no splitting into two open pieces	space_connected
nonempty	X has a point	space_nonempty
real dimension 6	$\text{finrank } \mathbb{R} (\mathbb{C} \times \mathbb{C}^2) = 6$	real_dimension
simply connected	$\pi_1(X) = 1$	layer6_simplyConnected
homology vanishes above 6	$H_n(X; \mathbb{Z}) = 0$ for $n > 6$	layer7_homology_vanishes_above_six
H_2 vanishes	$H_2(X; \mathbb{Z}) = 0$	layer7_homologyTwo_subsingleton

Why this list, and no more

These are the only facts about X used downstream. Steps 2 and 3 are stated for an arbitrary space with these properties: a compact, Hausdorff, second countable, simply connected smooth 6-manifold whose homology is that of S^6 . That is what makes the recognition step reusable, and what makes the dependency of the final theorem on the construction a short, checkable list.

Reading the table

The first seven rows are the geometry of the construction; the last three are the algebraic topology of step 2, listed here because they are also stated as properties of X. Every name is an alias in HopfMissing.Layers, so the statement each one abbreviates is exactly the statement proved in the extracted development.

Figure 06 of 18 · step 1c, the interface

layers L5 - L7 of the development

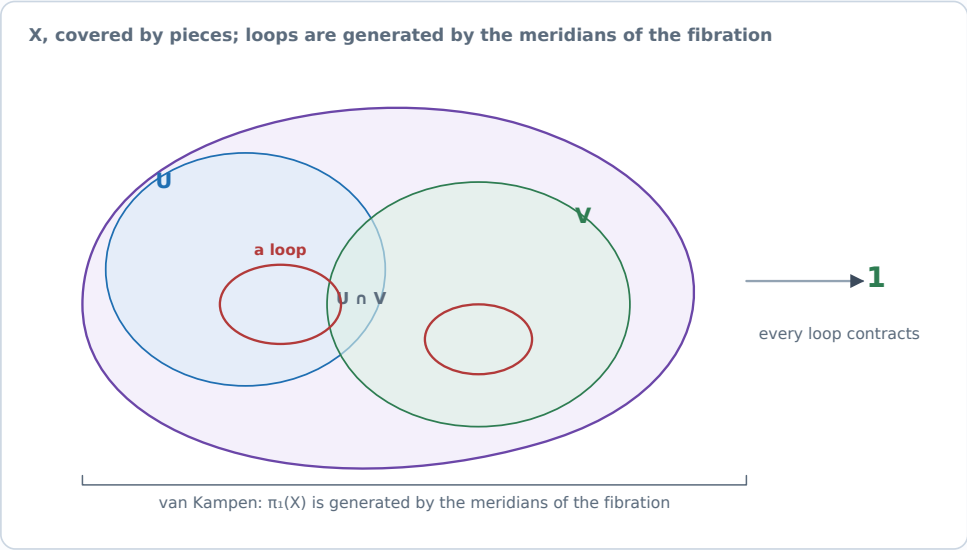
Every fact about the threefold that the rest of the proof consumes.

06-interface Figure 06 of 18 · step 1c, the interface

STEP 2a — THE FUNDAMENTAL GROUP $\pi_1(X) = 1$

STEP 2a — THE FUNDAMENTAL GROUP
 $\pi_1(X) = 1$

van Kampen, then every generator is killed



The argument

1. Cover X by the pieces of the fibration and apply van Kampen: $\pi_1(X)$ has a presentation whose generators are the meridian loops around the special fibres.
2. The monodromy of the family imposes relations on those generators — the elliptic power relations.
3. Those relations force every element of the group to be 1.

In Lean

`SpecialPeriods.Threefold.PiOne.trivial`: every element of the presented group is 1; with path-connectedness this gives `SpecialPeriods.Threefold.space_simplyConnected`, aliased as `layer6_simplyConnected`. This project also carries its own π_1 elimination and van Kampen infrastructure, which is what the comparison in `hopf-review/COMPARISON.md` is about.

Why π_1 matters here

Simple connectedness is used twice: to make Hurewicz available in step 2b, and as a hypothesis of the recognition theorem in step 3.

The Lean names

`SpecialPeriods.Threefold.space_simplyConnected` · `SpecialPeriods.Threefold.PiOne.trivial` · `layer6_simplyConnected`

STEP 2b — THE HOMOLOGY Singular homology, degree by degree

STEP 2b — THE HOMOLOGY

Singular homology, degree by degree

Mayer-Vietoris, Hurewicz, and the homology of the spheres

The homology of the six-sphere, computed in this repository:

k	0	1	2	3	4	5	6	≥ 7
$H_k(S^6; \mathbb{Z})$	$\mathbb{Z}$	0	0	0	0	0	$\mathbb{Z}$	0

What is proved about X before the comparison is made:

statement	content	Lean name
$\pi_1(X) = 1$	step 2a	layer6_simplyConnected
$H_n(X; \mathbb{Z}) = 0$ for $n > 6$	nothing above the top degree	layer7_homology_vanishes_above_six
$H_2(X; \mathbb{Z}) = 0$	the degree where the fibration could contribute	layer7_homologyTwo_subsingleton
$H_k(S^{n+1}; \mathbb{Z}) = 0$, $k \neq 0, n+1$	the sphere side of the comparison	layer2_sphere_homology_subsingleton

How the homology of X is computed

The space is assembled from pieces — period tori, mapping tori, cusp neighbourhoods — and the homology is propagated through the assembly by the Mayer-Vietoris sequence of a cover:

$$\dots \rightarrow H_k(A \cap B) \rightarrow H_k(A) \oplus H_k(B) \rightarrow H_k(A \cup B) \rightarrow H_{k-1}(A \cap B) \rightarrow \dots$$

Exactness turns knowledge of the pieces into knowledge of the union; the Hurewicz theorem converts the vanishing of low homotopy into homology.

The same theory, developed independently here

The singular-homology apparatus this argument needs — the long exact sequence of a pair, excision, Mayer-Vietoris, additivity, $H_0 \cong \mathbb{Z}[\pi_0]$, the degree-one Hurewicz isomorphism, and $H_m(S^n; \mathbb{Z})$ in every degree — is also built from scratch in this repository, in RequestProject/SphereHomology.lean and the modules it imports. Figure 16 is about that development; it is what contribution/ offers back to the reviewed repository.

Figure 08 of 18 · step 2b, the homology

layers L2 and L7 of the development

The homology of the six-sphere, and the homology of the threefold.

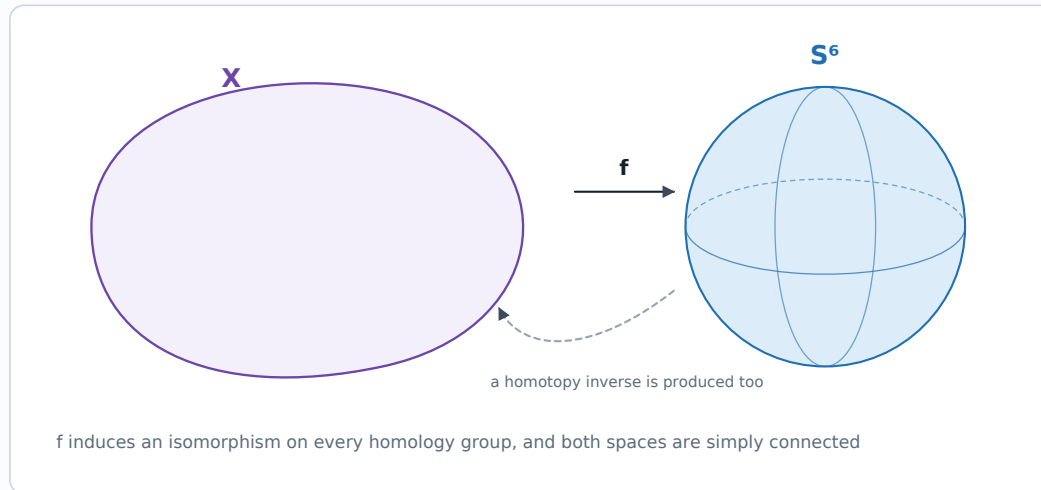
08-homology Figure 08 of 18 · step 2b, the homology

STEP 2c — THE HOMOTOPY EQUIVALENCE X and S^6 have the same homotopy type

STEP 2c — THE HOMOTOPY EQUIVALENCE

X and S^6 have the same homotopy type

Degree.threefoldHomotopyEquiv



Why an equivalence

A map between simply connected spaces that induces isomorphisms on all homology groups is a homotopy equivalence (Whitehead, via Hurewicz). Both sides are simply connected — X by step 2a, S^6 because it is a sphere of dimension ≥ 2 — and the homology matches by step 2b, so f is an equivalence.

In Lean

`Degree.threefoldHomotopyEquiv : Space  $\approx_h$   $S^6$` ,
aliased as `layer8_homotopyEquiv`. It is built from
`Degree.sphereHomotopyEquiv` at a chosen basepoint,
whose inverse is produced by a right-homotopy-inverse
argument.

What has been achieved, and what has not

X is now known to be homotopy equivalent to S^6 . That is far weaker than being S^6 : homotopy equivalence does not by itself give a homeomorphism. Step 3 supplies one.

What step 3 will use

X compact, Hausdorff, second countable, a smooth 6-manifold, and homotopy equivalent to S^6 — exactly the hypotheses of the recognition theorem.

The Lean names

`Degree.threefoldHomotopyEquiv` · `Degree.sphereHomotopyEquiv` · `layer8_homotopyEquiv`

Figure 09 of 18 · step 2c, the homotopy equivalence

layer L8 of the development

The homotopy equivalence between the threefold and the six-sphere.

09-homotopy-equivalence Figure 09 of 18 · step 2c, the homotopy equivalence

STEP 3a — MORSE THEORY Cancelling the critical points of a Morse function on X

STEP 3a — MORSE THEORY

Cancelling the critical points of a Morse function on X

MorseCancel.exists_two_critical_point_morse_of_homotopySixSphere



Figure 10 of 18 · step 3a, Morse theory layer L9 of the development

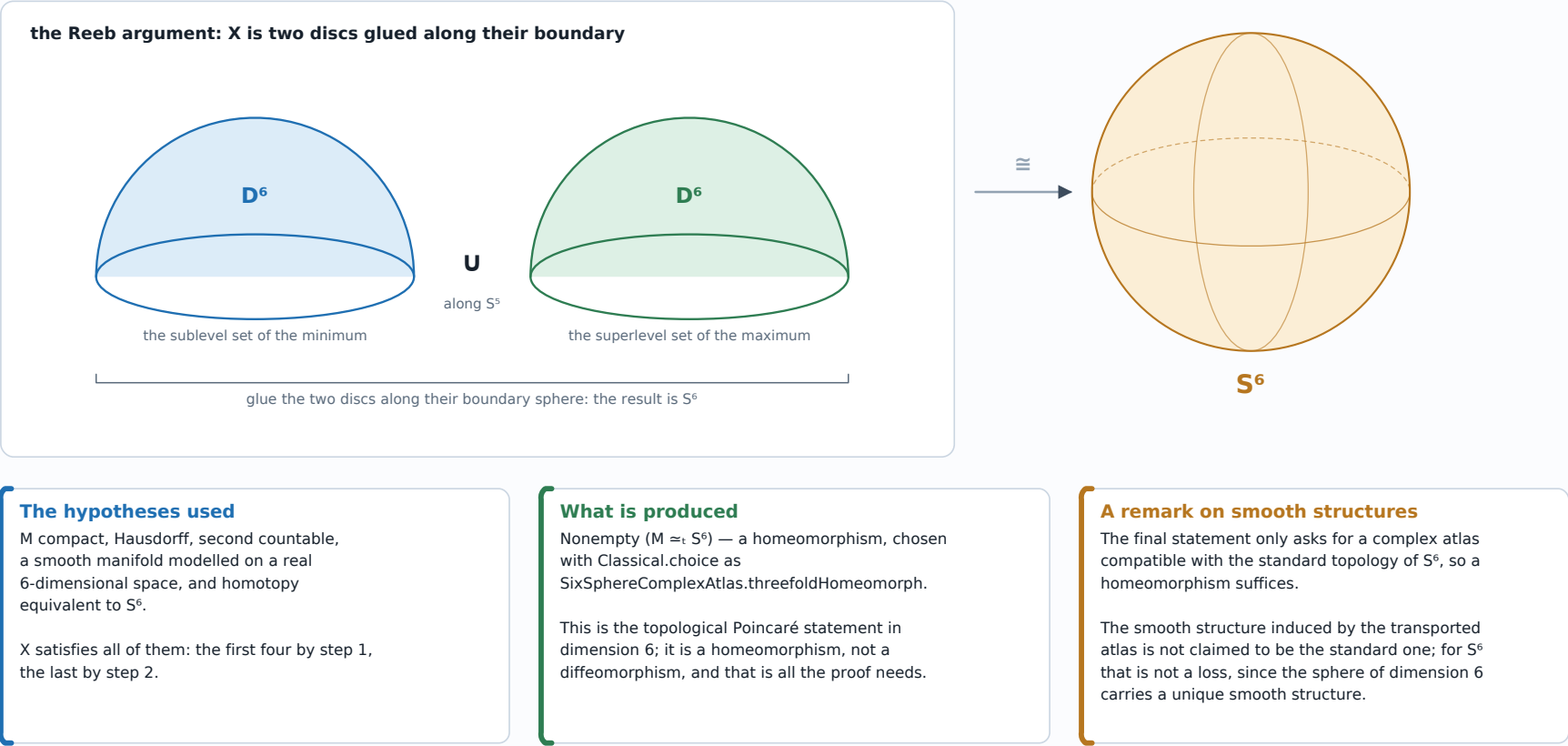
An ordered Morse function, and the cancellation of critical points. 10-morse Figure 10 of 18 · step 3a, Morse theory

STEP 3b — THE RECOGNITION Two critical points $\Rightarrow$ homeomorphic to S^6

STEP 3b — THE RECOGNITION

Two critical points $\Rightarrow$ homeomorphic to S^6

Smale.homeomorphic_sixSphere_of_homotopySixSphere



Two discs glued along a five-sphere, hence a homeomorphism.

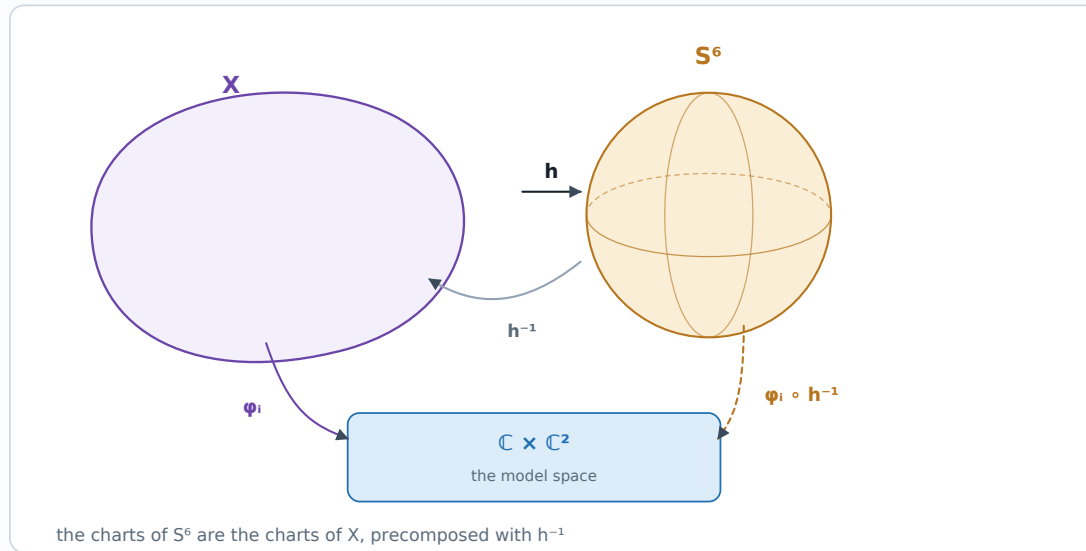
11-recognition Figure 11 of 18 · step 3b, the recognition

STEP 4a — TRANSPORT Moving the holomorphic atlas from X to S^6

STEP 4a — TRANSPORT

Moving the holomorphic atlas from X to S^6

ManifoldAtlasTransport.chartedSpace / .isManifold



Why the transported atlas is still holomorphic

The transition map of two transported charts is

$$(\varphi_i \circ h^{-1})^{-1} \gg (\varphi_j \circ h^{-1}) = \varphi_i^{-1} \gg \varphi_j,$$

literally the same partial map as before: h and h^{-1} cancel. So membership in the analytic groupoid is inherited, with no analysis to redo. In Lean this is `ManifoldAtlasTransport.transition_eq`.

What is used about h

Only that it is a homeomorphism: charts are moved along an open embedding, which is what `lift_openEmbedding` does. No smoothness of h is needed, and none is available — step 3 produces a homeomorphism only.

Step 4a in one line

A charted space structure on S^6 modelled on $\mathbb{C} \times \mathbb{C}^2$, whose transition maps are the transition maps of X , hence complex analytic.

The Lean names

`ManifoldAtlasTransport.chartedSpace` · `ManifoldAtlasTransport.transition_eq` · `ManifoldAtlasTransport.isManifold` · `SixSphereComplexAtlas.threefoldHomeomorph`

Figure 12 of 18 · step 4a, transport

the atlas is carried across the homeomorphism

The atlas transported, with the transition maps unchanged.

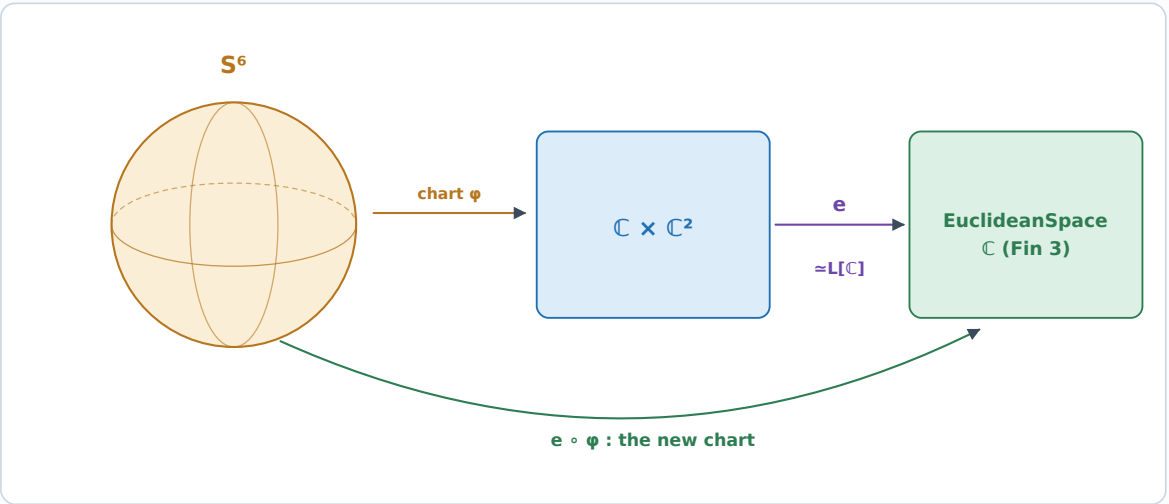
12-transport Figure 12 of 18 · step 4a, transport

STEP 4b — THE MODEL SPACE From $\mathbb{C} \times \mathbb{C}^2$ to EuclideanSpace $\mathbb{C}$ (Fin 3)

STEP 4b — THE MODEL SPACE

From $\mathbb{C} \times \mathbb{C}^2$ to EuclideanSpace $\mathbb{C}$ (Fin 3)

SpecialPeriods.Threefold.ModelChange



Why this is harmless

e is a continuous $\mathbb{C}$ -linear equivalence, so both e and e^{-1} are complex analytic. Composing every chart with e changes the model but not the transitions up to composition with e and e^{-1} , and the analytic groupoid is closed under that. The regularity ω is preserved.

Why it is needed

The target statement names one specific model, EuclideanSpace $\mathbb{C}$ (Fin 3). The construction naturally produces $\mathbb{C} \times \mathbb{C}^2$. The two are the same up to a linear isomorphism, and the statement must be met literally.

After step 4b

S^6 carries a ChartedSpace structure modelled on EuclideanSpace $\mathbb{C}$ (Fin 3) with complex analytic transition maps — which is the target proposition.

The Lean names

SpecialPeriods.Threefold.ModelChange.chartedSpace · ModelChange.isManifold · SixSphereComplexAtlas.modelEquiv · SixSphereComplexAtlas.exists_complex_analytic_atlas

Figure 13 of 18 · step 4b, the model space

the last adjustment before the statement is met

The model space changed to the one the target statement names.

13-model-change Figure 13 of 18 · step 4b, the model space

THE CONCLUSION The six-sphere carries a complex structure

THE CONCLUSION

The six-sphere carries a complex structure

and the two forms of the statement follow from each other

the theorem, as proved

```
theorem SixSphereComplexAtlas.exists_complex_analytic_atlas :  
  ∃ atlas : ChartedSpace (EuclideanSpace ℂ (Fin 3)) (unitSphere 6),  
    letI := atlas  
    IsManifold (modelWithCornersSelf ℂ (EuclideanSpace ℂ (Fin 3))) ω  
      (unitSphere 6)
```

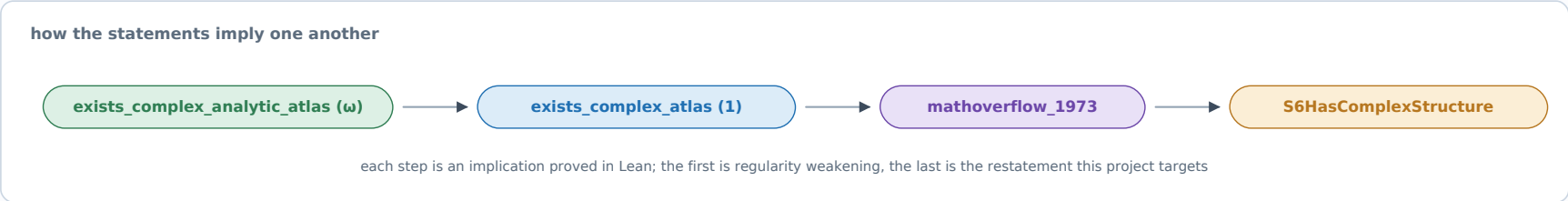
and the challenge form, with regularity 1, follows from it

The axiom closure

#print axioms reports, for every headline result of this repository:

[propext, Classical.choice, Quot.sound]

No sorry, no axiom declaration, no native_decide and no implemented_by attribute anywhere in the shipped sources. Figure 17 is about how that is checked.



What has been proved

There is an atlas of the standard six-sphere by open subsets of $\mathbb{C}^3$ whose transition maps are complex analytic. S^6 carries an integrable complex structure.

What has not been proved

Nothing is claimed about the induced smooth structure being standard, about uniqueness, or about the informal preprint matching the Lean development line by line.

WHAT THIS REPOSITORY ADDS (1) Several complex variables, built from scratch

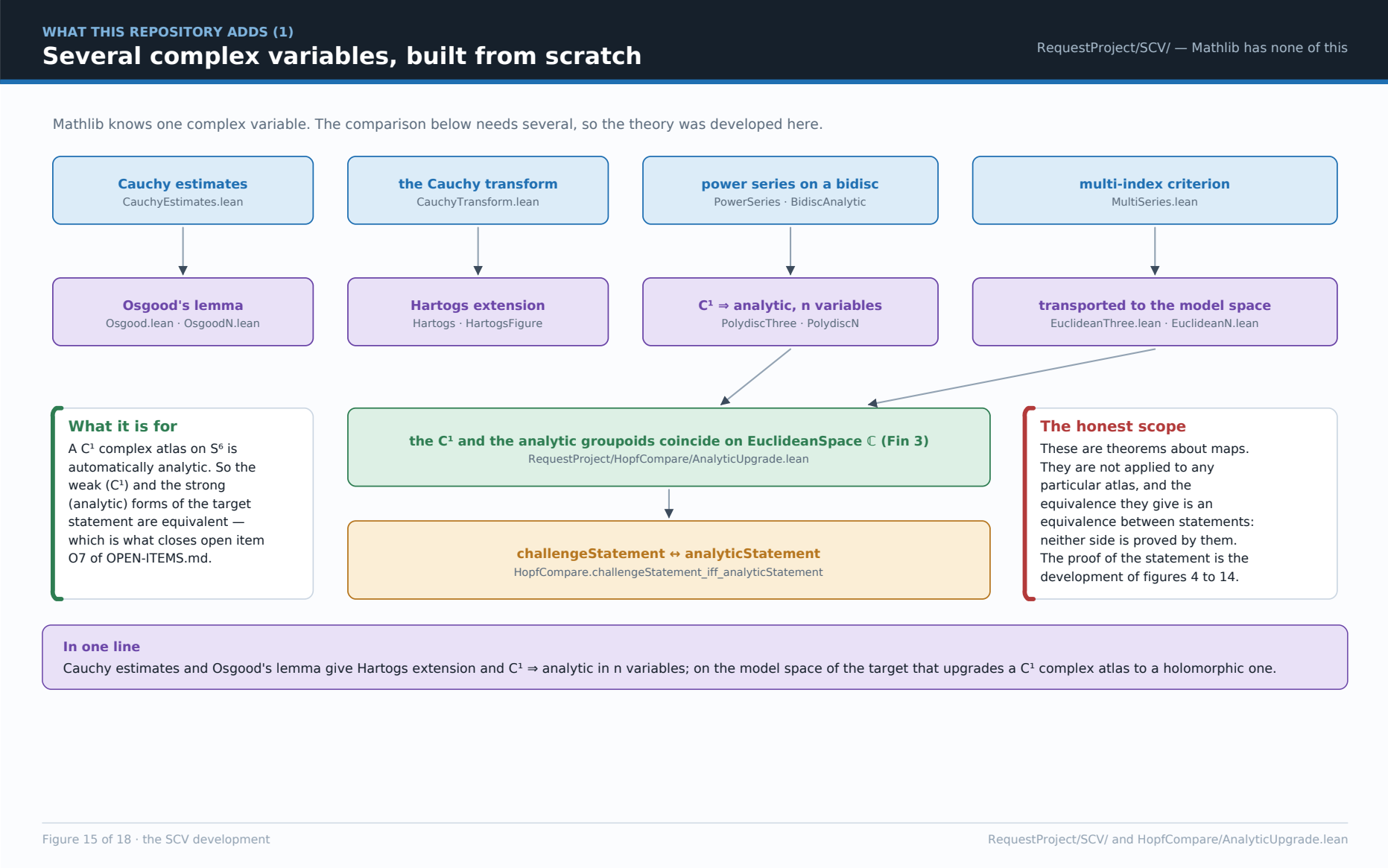


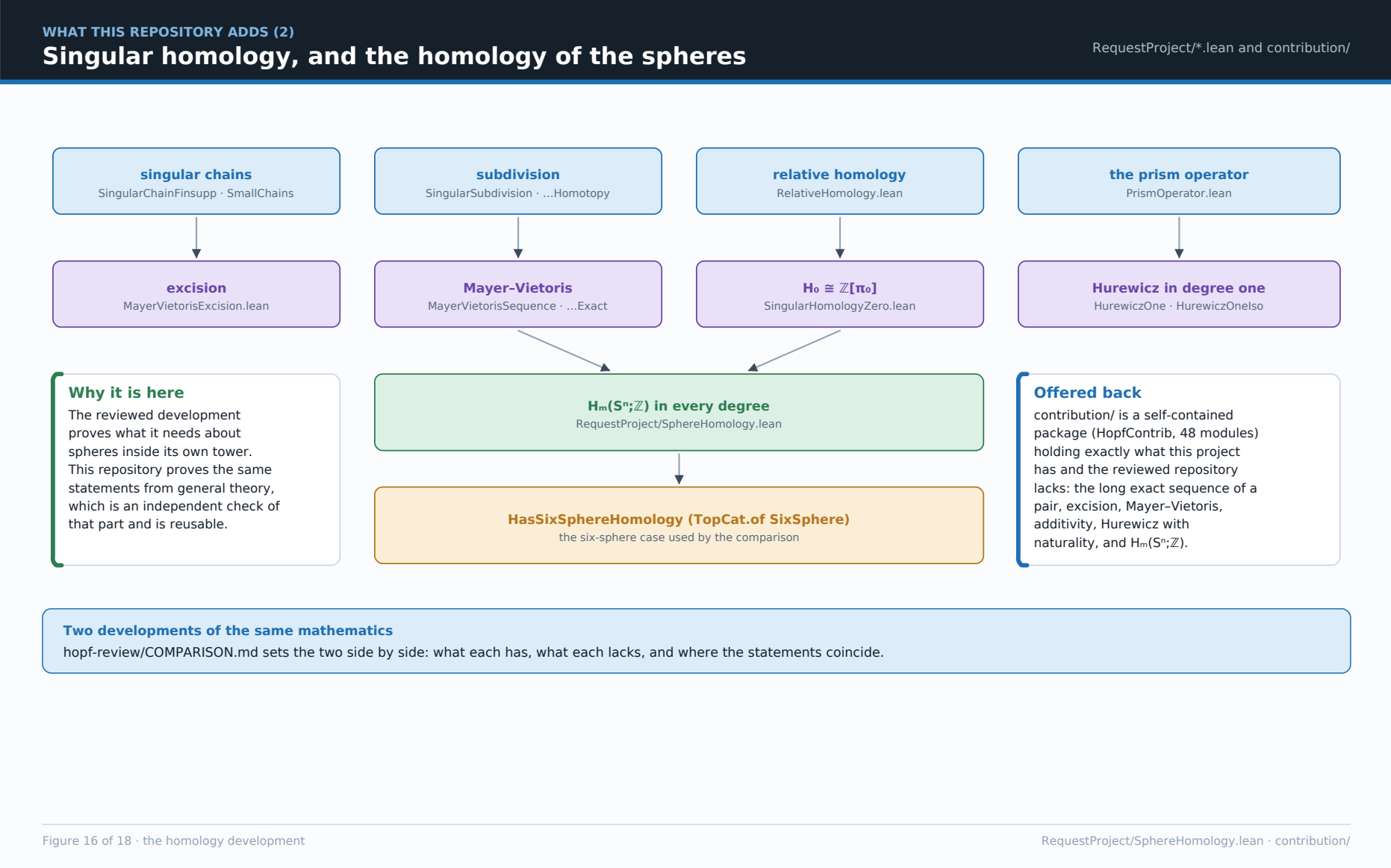
Figure 15 of 18 · the SCV development

RequestProject/SCV/ and HopfCompare/AnalyticUpgrade.lean

The several-complex-variables theory developed here.

15-several-complex-variables Figure 15 of 18 · the SCV development

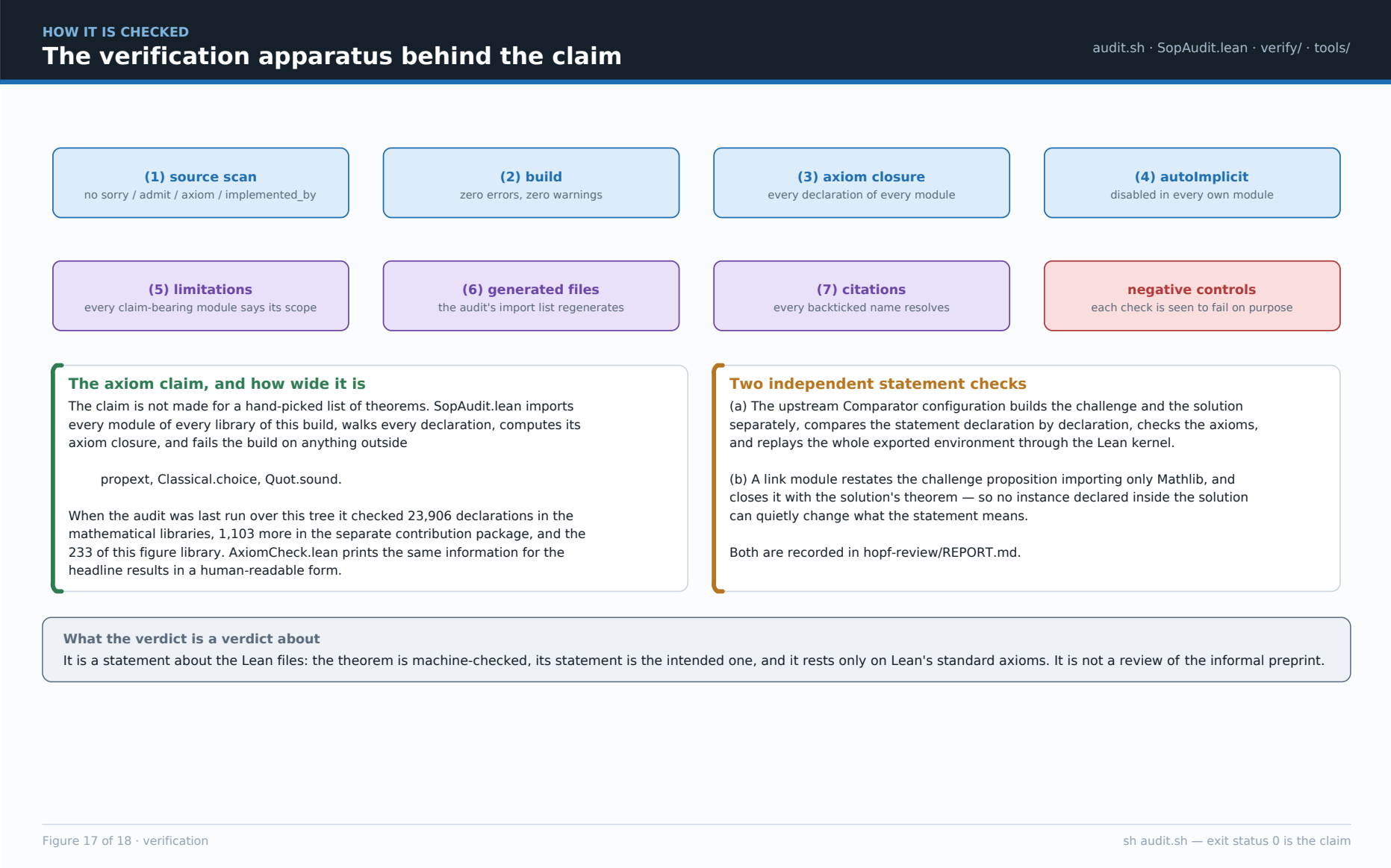
WHAT THIS REPOSITORY ADDS (2) Singular homology, and the homology of the spheres



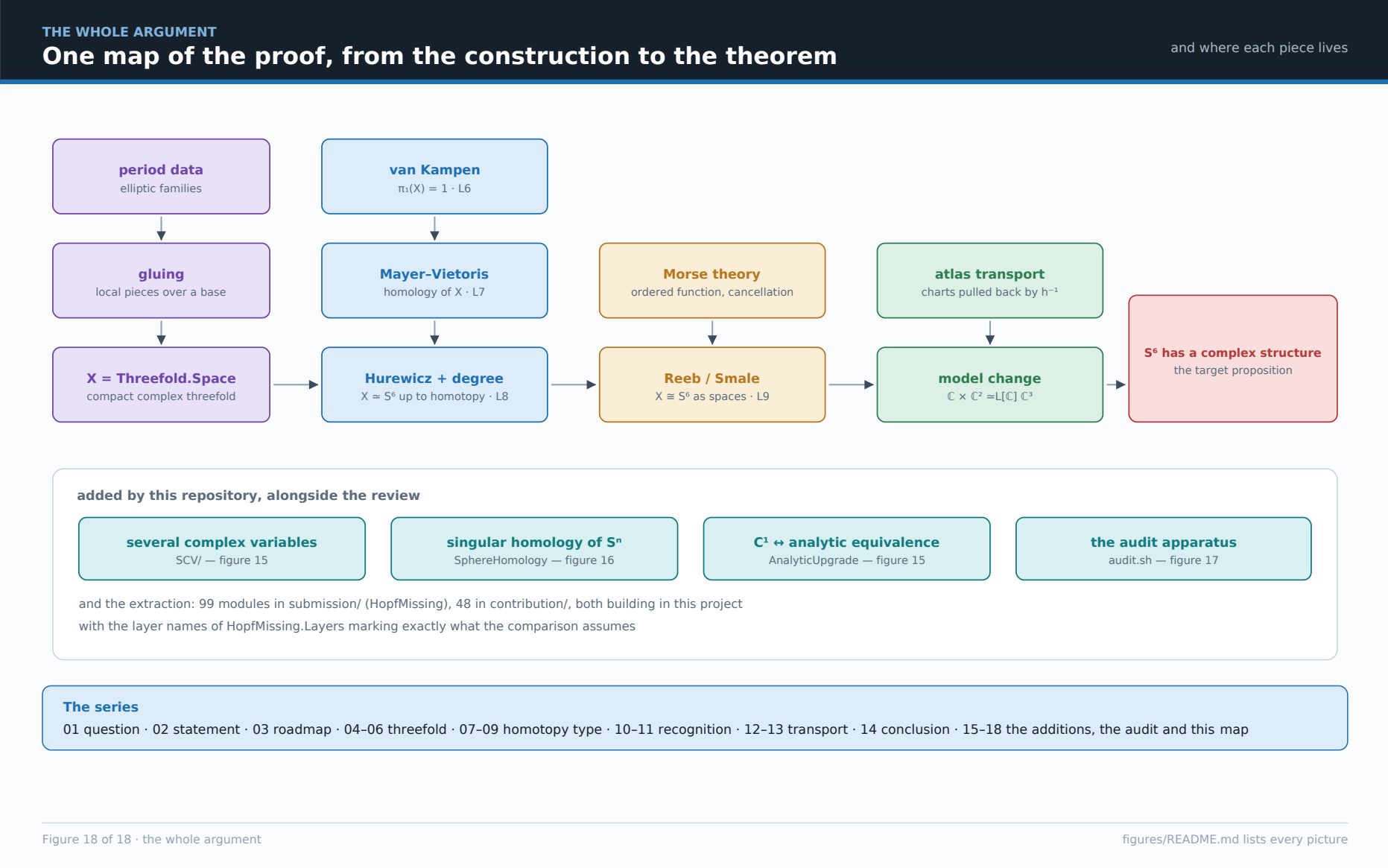
The singular-homology development here, and the packaged contribution.

16-homology-development Figure 16 of 18 · the homology development

HOW IT IS CHECKED The verification apparatus behind the claim



THE WHOLE ARGUMENT One map of the proof, from the construction to the theorem



The whole argument on one page.

18-whole-argument Figure 18 of 18 · the whole argument